

FOOD DELIVERY ANALYTICS CHALLENGE

AI & Data Science Hackathon Project Report

1. Introduction

1.1 Project Objective

The objective of this project is to analyze a food delivery dataset using Python and Pandas. The analysis focuses on understanding delivery performance and identifying the factors that affect delivery time.

The project examines delivery time, delivery distance, traffic conditions, weather conditions, delivery-person information, and delivery speed. The main goal is to convert the raw delivery data into useful business insights that can help a food delivery company improve its operations.

No Machine Learning model was used because the hackathon specifically required data analysis rather than ML.

1.2 Dataset

The provided dataset contains food delivery records with information about:

- Delivery persons
- Delivery locations
- Restaurant locations
- Order dates and times
- Weather conditions
- Road traffic density
- Vehicle condition
- Type of order
- Type of vehicle
- Multiple deliveries
- Festival information
- City
- Delivery time
- Delivery distance

The original dataset contained approximately 38,964 delivery records.

1.3 Tools and Libraries

The following tools and Python libraries were used:

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Google Gemini API
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2. Task A — Load and Understand the Data

The dataset was loaded into Python using Pandas.

The dataset was inspected to understand its structure, column names, data types, missing values, and duplicate records.

The main columns include:

- ID
- Delivery_person_ID
- Delivery_person_Age
- Delivery_person_Ratings
- Restaurant_latitude
- Restaurant_longitude
- Delivery_location_latitude
- Delivery_location_longitude
- Order_Date
- Time_Orderd
- Time_Order_picked
- Weather_conditions
- Road_traffic_density
- Vehicle_condition
- Type_of_order
- Type_of_vehicle
- multiple_deliveries
- Festival
- City
- Time_taken (min)
- distance_km
- delivery_speed

Duplicate Records

The dataset was checked for duplicate rows.

Result:

Duplicate records = 0

Therefore, no duplicate rows needed to be removed.

3. Task B — Data Cleaning

The dataset contained missing values in some columns.

Missing values were found mainly in:

- Delivery_person_Age
- Delivery_person_Ratings
- Time_Orderd

The missing values were handled using suitable preprocessing methods.

Rows with missing order-time information were removed because the order time was not available and could affect time-related analysis.

Missing delivery-person age and rating values were replaced using their respective median values. Median was selected because it is less affected by extreme values.

After cleaning, the required analysis columns contained no missing values.

Cleaning Results

- Missing values after cleaning = 0
- Duplicate records = 0
- Records available for analysis = 38,129

The cleaned dataset was then used for the remaining analysis.

4. Task C — Basic Analysis

After cleaning the dataset, basic statistics were calculated using Pandas.

4.1 Total Deliveries

Total number of deliveries analyzed:

38,129 deliveries

4.2 Delivery Time

The delivery-time statistics were:

- Average delivery time: 26.58 minutes
- Minimum delivery time: 10 minutes
- Maximum delivery time: 54 minutes

This shows that delivery times varied considerably across different orders.

4.3 Average Delivery Distance

The average delivery distance was:

9.76 km

This represents the average distance traveled for a delivery in the cleaned dataset.

4.4 Average Delivery Speed

The numeric delivery speed was calculated using delivery distance and delivery time.

Average delivery speed:

23.58 km/h

The calculation was based on:

Delivery Speed = Distance / Delivery Time

The time was converted from minutes to hours to obtain speed in km/h.

4.5 Delivery-Person Rating

The average delivery-person rating was:

4.63

This indicates that delivery persons generally received high customer ratings.

4.6 Delivery-Person Age

The average delivery-person age was:

29.61 years

This represents the average age of the delivery persons in the cleaned dataset.

5. Task D — Competition Questions

Q1 — Traffic Impact

Question

Which road traffic condition has the highest average delivery time?

The average delivery time was grouped according to road traffic density.

The result showed that:

Highest traffic condition = Jam

Average delivery time during Jam = 31.42 minutes

Conclusion

Road traffic has a significant effect on delivery performance. Deliveries during traffic jams take longer than deliveries under less congested road conditions.

This suggests that traffic congestion is an important factor that the food delivery company should consider when estimating delivery times.

Q2 — Distance Impact

Question

How does delivery distance affect delivery time?

The relationship between delivery distance and delivery time was analyzed using correlation.

The calculated correlation was:

Distance-Time Correlation = 0.322

Conclusion

The correlation is positive, which means that longer delivery distances generally tend to be associated with longer delivery times.

However, the correlation is not very strong. This suggests that distance alone does not explain most of the variation in delivery time.

Other factors, such as traffic, weather, and operational conditions, also influence delivery time.

Therefore, the company should not estimate delivery time based only on distance.

Q3 — Combined Conditions

Question

Which combination of weather condition and traffic density has the highest average delivery time?

Weather conditions and road traffic density were combined and analyzed together.

The highest average delivery time was found for:

Weather = Fog

Traffic = Jam

Average delivery time = 36.88 minutes

Conclusion

Fog combined with heavy traffic creates the most difficult delivery conditions.

Under these conditions, delivery times are considerably higher than the overall average delivery time of 26.58 minutes.

This indicates that the company should consider both weather and traffic together when estimating delivery times.

6. Task E — Data Visualizations

Two main visualizations were created to communicate the analysis results.

6.1 Average Delivery Time by Traffic Density

A bar chart was created to compare the average delivery time under different traffic conditions.

The chart helps identify which traffic condition causes the longest delivery times.

The analysis showed that Jam had the highest average delivery time at approximately 31.42 minutes.

Figure: Average Delivery Time by Traffic Density

[Insert traffic_density.png here]

6.2 Delivery Distance vs Delivery Time

A scatter plot was created to examine the relationship between delivery distance and delivery time.

The scatter plot supports the correlation analysis and shows that delivery time generally increases as distance increases, but the relationship is not extremely strong.

Figure: Delivery Distance vs Delivery Time

[Insert distance_vs_time.png here]

7. Task F — Business Insights

Based on the analysis, the following business insights were identified.

Insight 1 — Traffic Jams Increase Delivery Time

Deliveries during traffic jams had the highest average delivery time among the traffic conditions, at approximately 31.42 minutes.

Business Meaning

The company should consider traffic density when calculating estimated delivery times. During peak traffic hours, the system could provide customers with more realistic delivery estimates.

Insight 2 — Distance Is Not the Only Important Factor

The distance-time correlation was 0.322.

Business Meaning

Although longer distances generally increase delivery time, distance alone cannot accurately predict delivery performance. The company should also consider traffic, weather, and other operational conditions.

Insight 3 — Fog and Traffic Create the Worst Conditions

The combination of Fog and Jam produced the highest average delivery time of approximately 36.88 minutes.

Business Meaning

The company should pay special attention to deliveries made during poor weather and heavy traffic. Delivery estimates can be adjusted when these conditions occur.

8. Task G — AI-Powered Explanation

Google Gemini was used to convert the calculated analytical results into a simple business explanation.

The Python analysis was performed first using Pandas. The calculated results were then provided to the Gemini model.

The AI was not used to perform the primary statistical analysis. Instead, it was used to explain the already-calculated results in business language and provide practical recommendations.

AI-Generated Business Explanation

The analysis of 38,129 deliveries shows that external conditions such as traffic and weather can have a noticeable effect on delivery performance.

Traffic jams produced the highest average delivery time at approximately 31.42 minutes. The distance-time correlation of 0.322 shows a positive but moderate relationship between distance and delivery time, meaning that distance affects delivery time but is not the only factor.

The most difficult combination was Fog and Jam, where the average delivery time reached approximately 36.88 minutes.

These findings suggest that a food delivery company should use traffic and weather information along with delivery distance when estimating delivery times.

Practical Recommendations

1. Adjust estimated delivery times during heavy traffic and poor weather conditions.
 2. Monitor weather and traffic conditions together because their combination can cause significantly longer delivery times.
 3. Use traffic-aware delivery planning and consider assigning suitable vehicles to areas experiencing heavy congestion.
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9. Project Workflow

The complete project followed the required workflow:

Load → Clean → Analyze → Visualize → Interpret → Explain

First, the dataset was loaded and inspected. Missing and duplicate values were investigated and cleaned. Basic statistics were then calculated using Pandas.

After that, the three competition questions were answered programmatically. Visualizations were created to communicate the main findings. Finally, business insights were developed and Gemini was used to provide a simple business explanation.

10. Final Conclusion

The food delivery analysis shows that delivery performance is influenced by several factors.

The dataset contained 38,129 usable delivery records after cleaning. The average delivery time was 26.58 minutes, while the average delivery distance was 9.76 km and the average calculated delivery speed was 23.58 km/h.

Traffic congestion was an important factor, with Jam having the highest average delivery time of 31.42 minutes. Distance had a positive but moderate relationship with delivery time, with a correlation of 0.322.

The most challenging combination was Fog and Jam, which resulted in an average delivery time of 36.88 minutes.

Overall, the analysis demonstrates that food delivery companies can improve delivery estimates and operational planning by considering multiple factors such as distance, traffic, and weather rather than relying on distance alone.

The project successfully demonstrates how Python and data analysis can be used to convert raw delivery data into practical business decisions.